

Lecture 8

PARALLELISM BASICS

Adapted from Stanford CS336

<https://cs336.stanford.edu/>

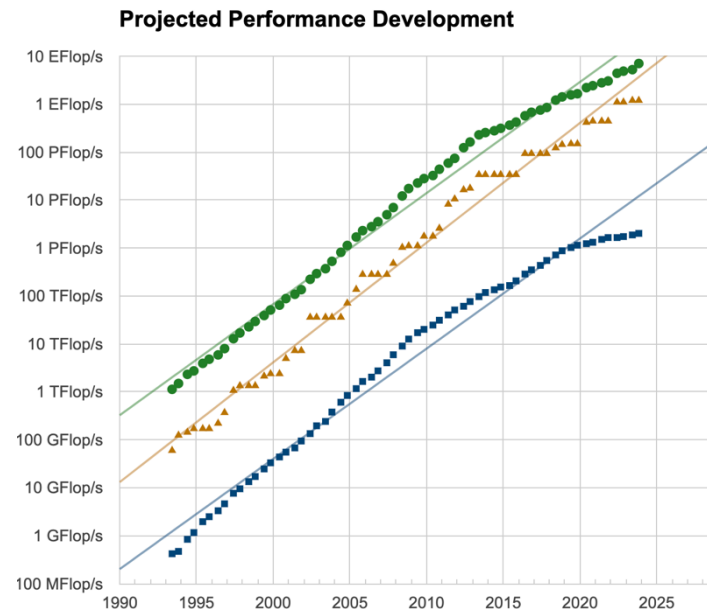
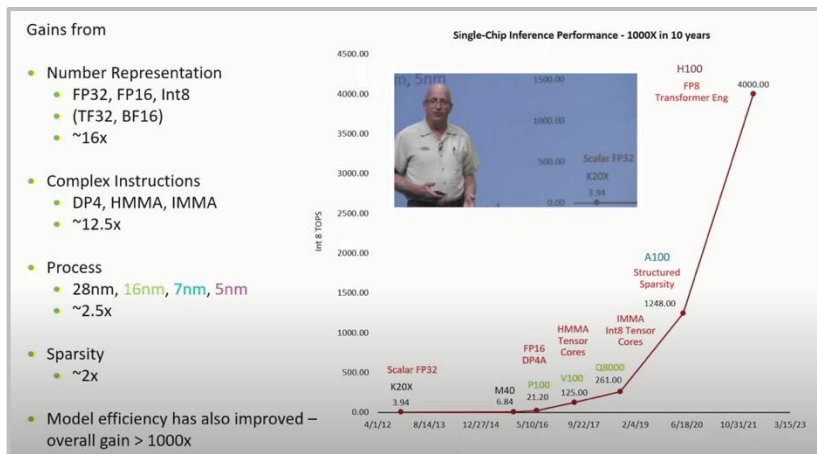
Outline and goals

- Understand the systems complexities of training huge models
- Different parallelization paradigms and why people use multiple at once
- What large scale training runs often look like

Organization today:

- ❖ **Part 1:** Basics of networking for LLMs
- ❖ **Part 2:** Different forms of parallel LLM training
- ❖ **Part 3:** Scaling and training big LMs with parallelism

Limits to GPU-based scaling – compute

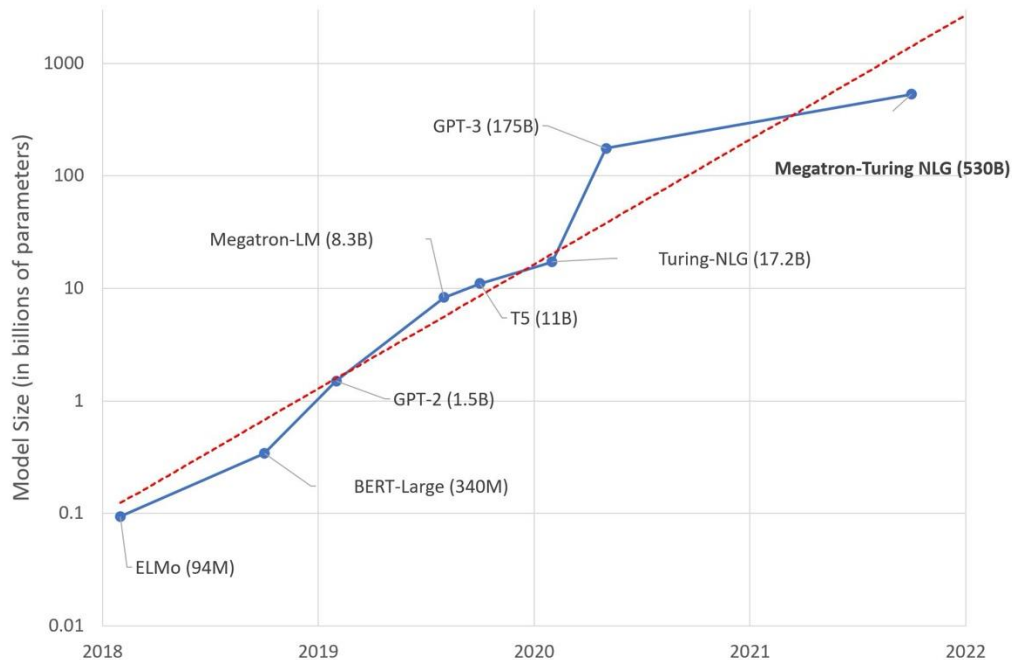


There are limits to single-GPU scaling.

The world's fastest supercomputers have *exaflops* of compute

Limits to GPU-based scaling - memory

Models are getting really big..

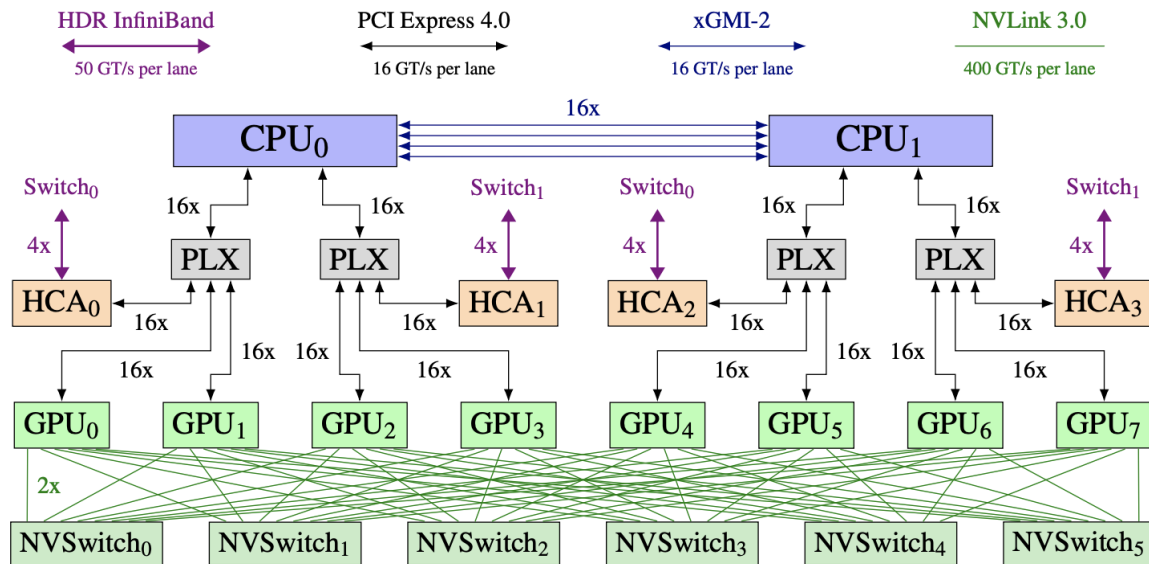


A single GPU can't fit most of these large models!

What do we do? Multi-GPU, multi-machine parallelism

Intra-node parallelism
via high-speed interconnects

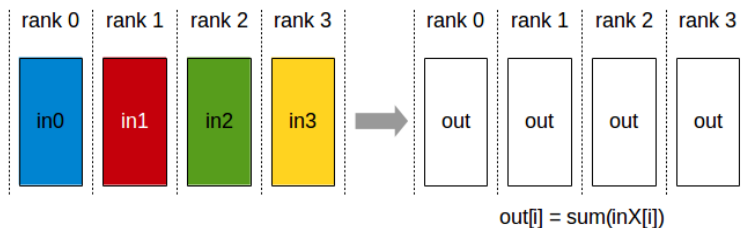
High-speed inter-node parallelism



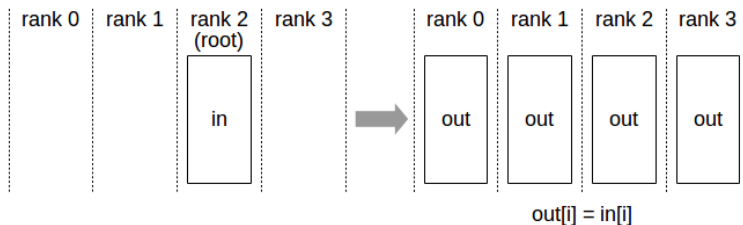
Split up memory and compute requirements across GPUs and machines

But first.. Some basics about collective communication

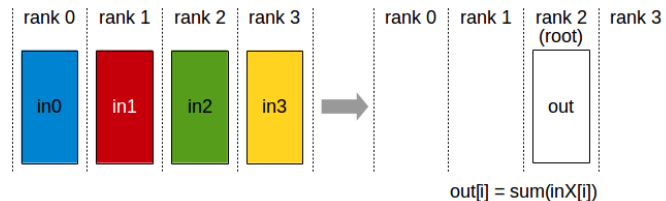
All reduce



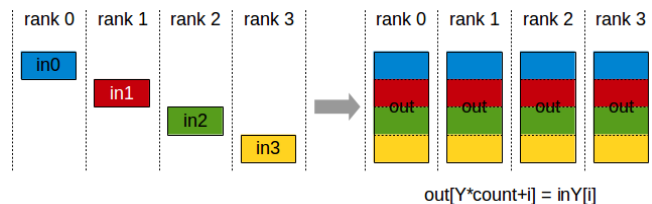
Broadcast



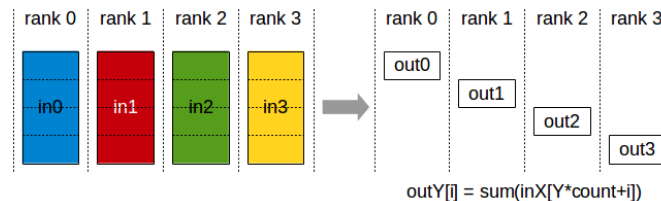
Reduce



All Gather

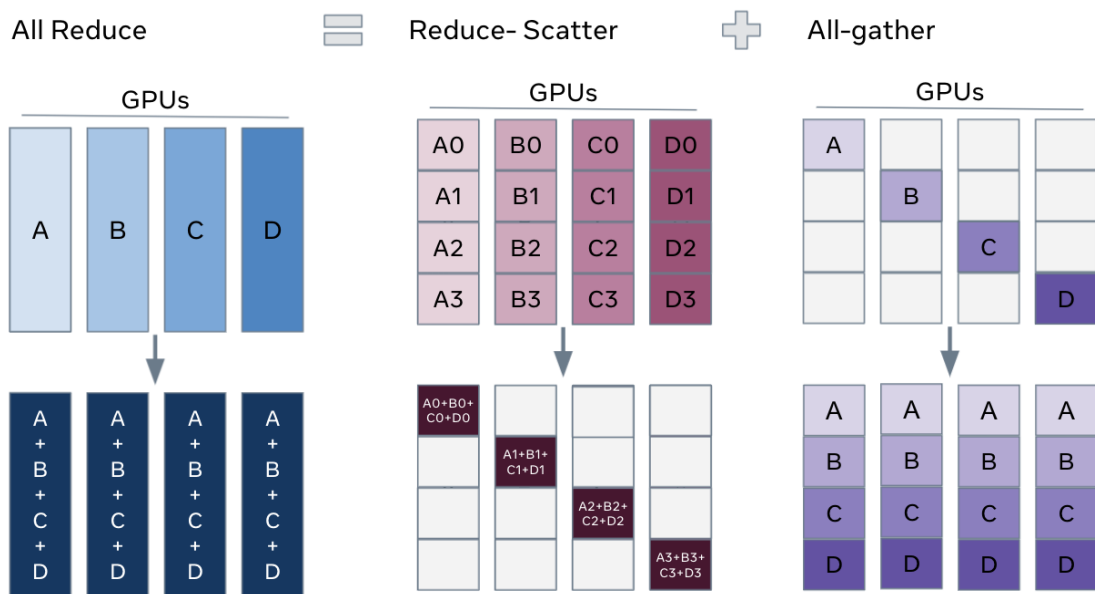


Reduce Scatter



Important detail – all reduce vs reduce-scatter-gather.

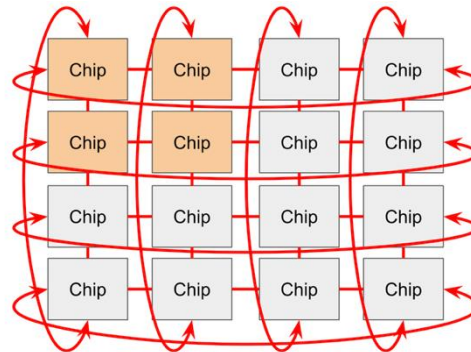
Reduce can be implemented as two steps: reduce-scatter and all-gather



Importantly, in the bandwidth-limited regime, this is the best you can do

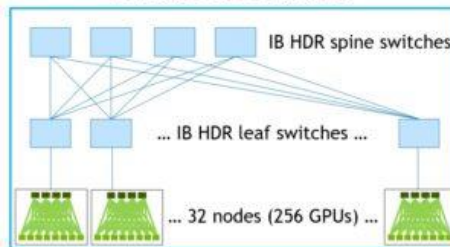
TPUs vs GPUs – design differences at the comm level

TPU networking
toroidal mesh

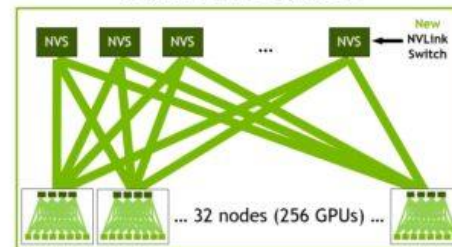


GPU networking
All-to-all (up to 256)

DGX A100 256 SuperPOD



DGX H100 256 SuperPOD



Fully NVLink-connected
Massive bisection bandwidth

	A100 SuperPod			H100 SuperPod			Speedup	
	Dense PFLOP/s	Bisection [GB/s]	Reduce [GB/s]	Dense PFLOP/s	Bisection [GB/s]	Reduce [GB/s]	Bisection	Reduce
1 DGX / 8 GPUs	2.5	2,400	150	16	3,600	450	1.5x	3x
32 DGXs / 256 GPUs	80	6,400	100	512	57,600	450	9x	4.5x

Mesh vs tree vs other

Why mesh? Why tree?

Pro mesh – Cheaper to operate, can be made fast (and just do tensor parallel)

Pro tree/A2A – Better for less structured communications (expert parallel)

taekim.substack.com/p/nvidias-bill-dally-and-googles-jeff

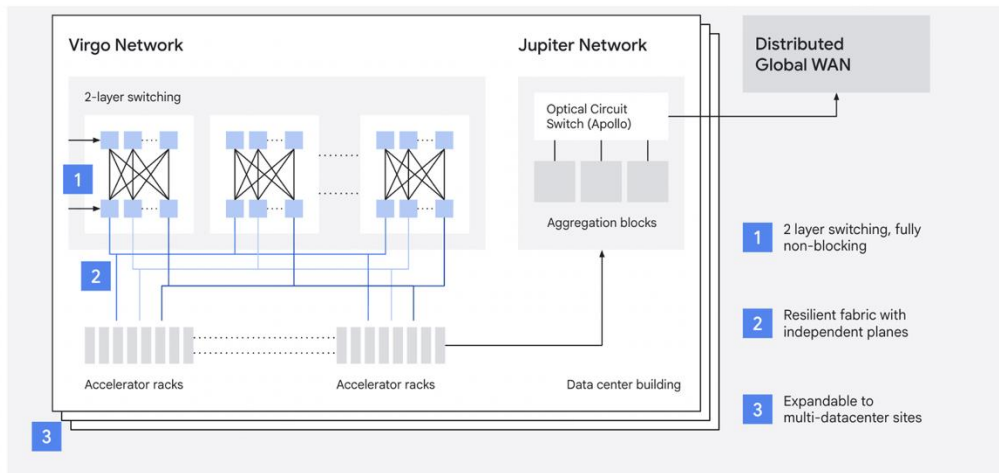
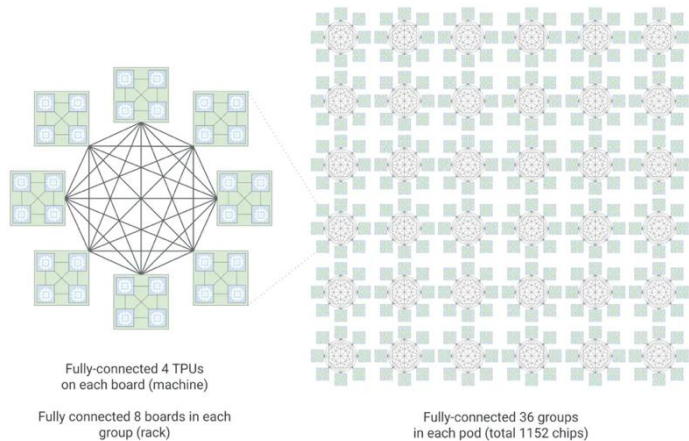
Dally and Dean on Google TPUs versus Nvidia GPUs. GPUs are good for ..

Mixture of Experts models and TPUs are good for local workloads

Dally: “You can’t say one is better than the other because it really has to do with what is the workload and what is the traffic pattern. If you have a very local workload, a direct connected network with relatively low radix like a 3D torus, is ideal because you can map your problem. So part of the problem is on this TPU and part of the problem is on this TPU and you have to go one hop to get your data over there. But then there are other cases where, for example, you have an MOE [mixture of experts] model and you’ve got a lot of experts, and they’re scattered all over the place and now you’re having to take many hops to get to a given expert. So it would be more efficient to do one hop up to a switch and one hop down to the expert that you want to be at” “Given a traffic pattern, you’d come up with an optimum network, but there’s no network that is good for all traffic patterns.”

Dean: “I totally agree. It depends on exactly what your workload is.”

But then things change.. TPU8i/t



TPU8i networking – closer to tree-style topologies (maybe for MoEs?)

TPU8t scale-out networking – ‘Virgo’ with switched networks

Why not connect everything?

Domain sizes – why not connect everything?

Key Capabilities - Huawei Ascend 910C Cloud Matrix 384 vs Nvidia GB200 NVL72				
Chip and Package Level				
	Unit	GB200	Ascend 910C	Huawei vs Nvidia
BF16 dense TFLOPS	TFLOPS	2,500	780	0.3x
HBM capacity	GB	192	128	0.7x
HBM bandwidth	TB/s	8.0	3.2	0.4x
Scale Up Bandwidth	Gb/s uni-di	7,200	2,800	0.4x
Scale Out Bandwidth	Gb/s uni-di	400	400	1.0x
System Level				
	Unit	Nvidia GB200 NVL72	Cloud Matrix CM384	Huawei vs Nvidia
BF16 dense PFLOPS	PFLOPS	180	300	1.7x
HBM capacity	TB	13.8	49.2	3.6x
HBM bandwidth	TB/s	576	1,229	2.1x
Scale Up Bandwidth	Gb/s uni-di	518,400	1,075,200	2.1x
Scale Up Domain Size	GPUs	72	384	5.3x
Scale Out Bandwidth	Gb/s uni-di	28,800	153,600	5.3x
All-In System Power ₁	W	145,000	599,821	4.1x
All-in Power per BF16 dense FLOP	W/TFLOP	0.81	2.00	2.5x
All-in Power per memory bandwidth	W per TB/s	251.7	488.1	1.9x
All-in Power per memory capacity	kW/TB	10.5	12.2	1.2x

1. All-in System Power is total cluster power including scale-out networking, storage, etc.



Part 1 recap

- ❖ New unit of compute – the datacenter
- ❖ What we want from multi-machine scaling:
 - ❖ Linear memory scaling (max model params scales with num gpus)
 - ❖ Linear compute scaling (model flops scale linearly with num gpus)
- ❖ Simple collective comms primitives

Part 2 – Standard LLM parallelization primitives

How do we parallelize LLMs? 3 important ideas

- **Data parallelism**
 - Naïve data parallel
 - ZeRO levels 1-3
- **Model parallelism**
 - Pipeline parallel
 - Tensor parallel
- **Activation parallelism**
 - Sequence parallel

Naïve data parallelism

Starting point – imagine we are doing naïve SGD

$$\theta_{t+1} = \theta_t - \eta \sum_{i=1}^B \nabla f(x_i)$$

Naïve parallelism: split the elements of B sized batch across M machines. Exchange gradients to synchronize

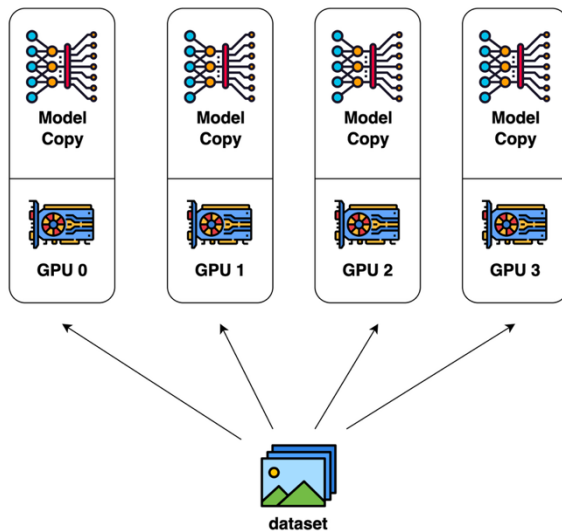
How does this do?

Compute scaling – each GPU gets B/M examples.

Communication overhead – transmits 2x # params every batch. OK if batches are big

Memory scaling – none. Every GPU needs # params at least

What's wrong with naïve data parallel?



Memory seems like it'd be a problem – we copy the model parameters to each GPU.
Let's take a closer look..

What's wrong with naïve data parallelism? - Memory

Our memory situation is actually *terrible*.

Depending on our precision..

We need 5 copies of weights and 16 bytes per param!

- 2 bytes for FP/BF 16 model parameters
- 2 bytes for FP/BF 16 gradients
- 4 bytes for FP32 master weights (the thing you accumulate into in SGD)
- 4 (or 2) bytes for FP32/BF16 Adam first moment estimates
- 4 (or 2) bytes for FP32/BF16 Adam second moment estimates

“Optimizer state”

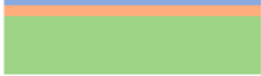
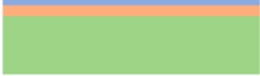
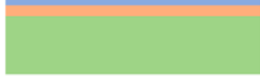
ZeRO – solving the memory overhead issue of DP

Core idea: split up the expensive parts (state) and use the reduce-scatter equivalence.

	gpu ₀ ... gpu _i ... gpu _{N-1}	Memory Consumed	K=12 $\Psi=7.5\text{B}$ $N_d=64$
Baseline		$(2 + 2 + K) * \Psi$	120GB
P _{os}		$2\Psi + 2\Psi + \frac{K * \Psi}{N_d}$	31.4GB
P _{os+g}		$2\Psi + \frac{(2 + K) * \Psi}{N_d}$	16.6GB
P _{os+g+p}		$\frac{(2 + 2 + K) * \Psi}{N_d}$	1.9GB

■ Parameters
 ■ Gradients
 ■ Optimizer States

ZeRO stage 1. optimizer state sharding

	gpu ₀	...	gpu _i	...	gpu _{N-1}	Memory Consumed	K=12 $\Psi=7.5\text{B}$ $N_d=64$
Baseline		...		...		$(2 + 2 + K) * \Psi$	120GB
P _{os}		...		...		$2\Psi + 2\Psi + \frac{K * \Psi}{N_d}$	31.4GB

High level idea:

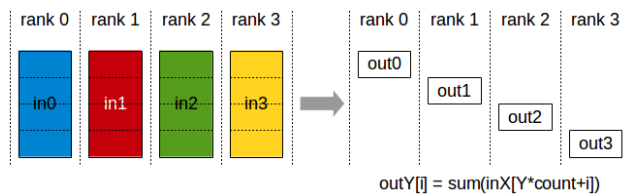
- Split up the optimizer state (first + second moments) across GPUs
- Everyone has the parameters + gradients

Each worker is responsible for updating a subset of params (corresponding to their slice)

ZeRO stage 1. how it works

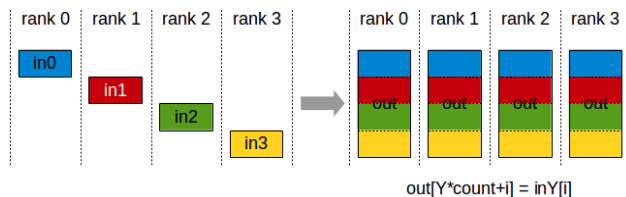
Step 1. Everyone computes a full gradient on their subset of the batch

Step 2. ReduceScatter the gradients – incur #params communication cost



Step 3. Each machine updates their param using their gradient + state.

Step 4. All Gather the parameters – incur #params communication cost



Comparing ZeRO stage 1 and naïve data parallel

	Naïve DDP	ZeRO stage 1
Communication primitive	One all-reduce (gradients)	One reduce scatter (send gradients) + all gather (collect params)
Communication cost	$2 * \# \text{ params}$	$2 * \# \text{ params}$
Memory	$(4+K) * \# \text{ params}$	$(4+K/N_{\text{gpu}}) * \# \text{ params}$

ZeRO stage 1 is *free* (in the bandwidth limited regime) memory wins

ZeRO stage 2. the simple extension to gradient sharding



Emboldened by our success, let's shard even more stuff

High level idea

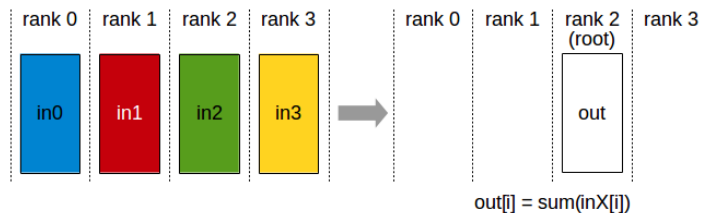
- Also keep the gradients (pink slices) sharded across the machines.
- Use the same (rough) tricks as stage 1.

Complexity – we can never instantiate a full gradient vector, but each worker must compute a full gradient (since we're data parallel)

ZeRO stage 2. how it works

Step 1. Everyone incrementally goes backward on the computation graph

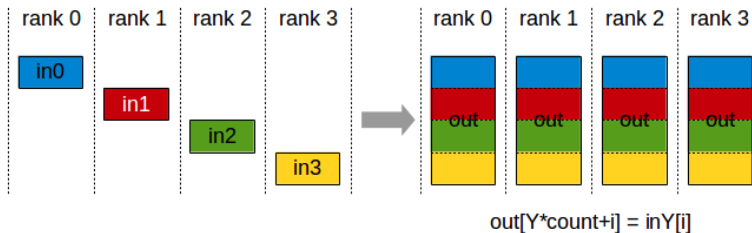
Step 1a. After computing a layer's gradients, immediately reduce to send this to the right worker



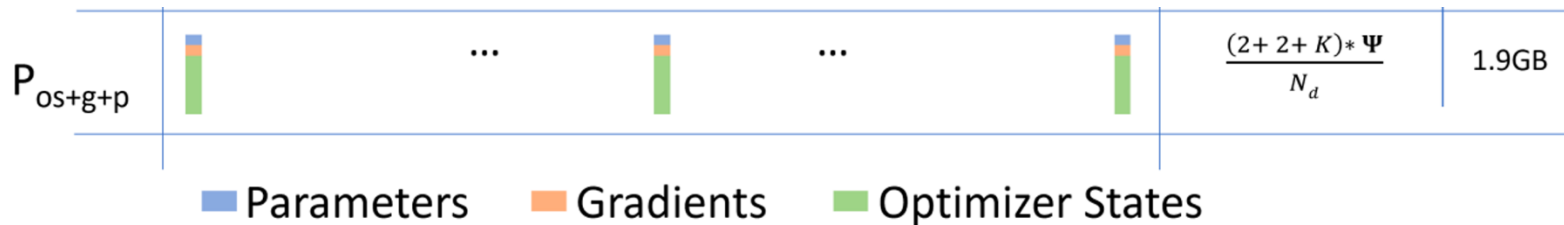
Step 1b. Once gradients are not needed in the backward graph, immediately free it.

Step 2. Each machine updates their param using their gradient + state.

Step 3. All Gather the parameters.



ZeRO stage 3 (aka FSDP) shard *everything*



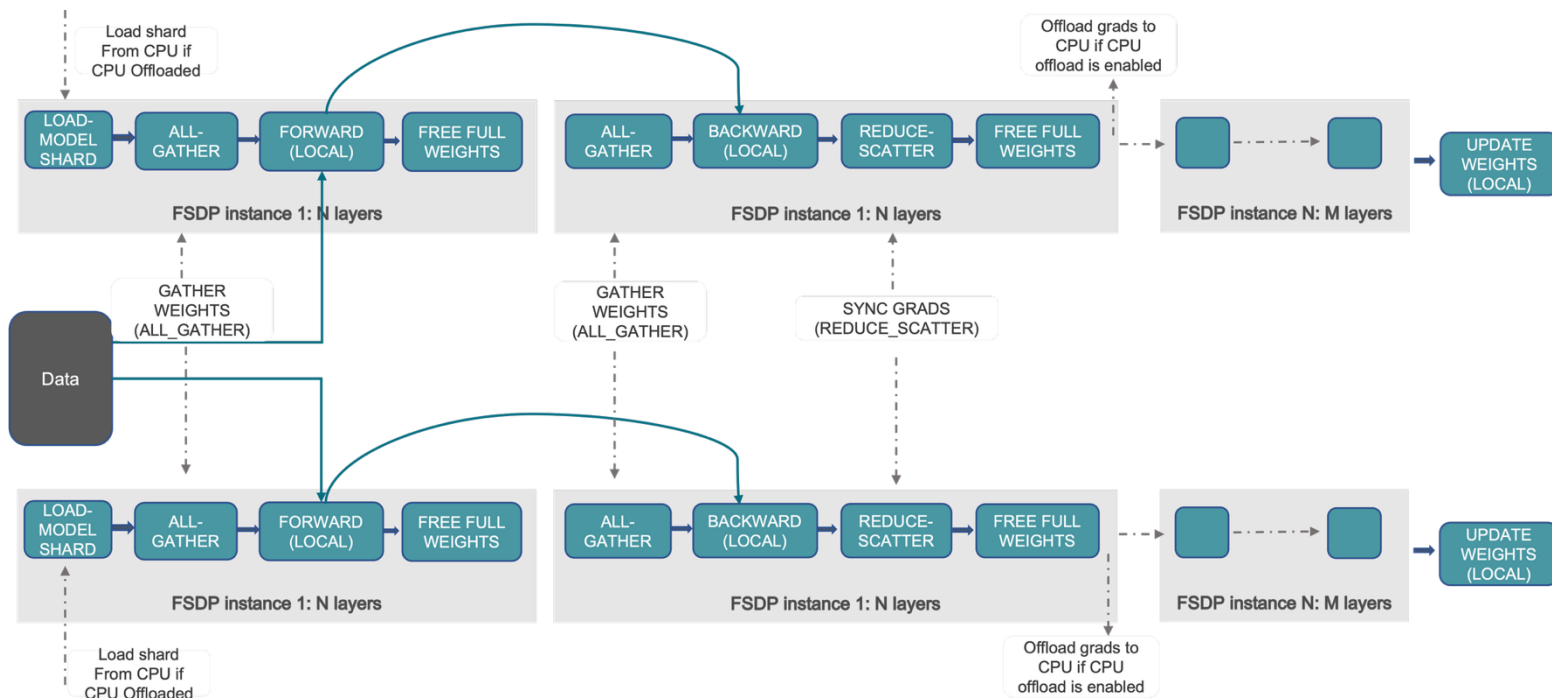
We've gotten almost everything for free so far.. lets try to solve *all* our memory issues

High level idea

- Shard everything – incl parameters!
- Use the same ‘incremental communication / computation’ ideas
- Send and request parameters on demand while stepping through the compute graph.

Is it possible to do this with low overhead?

ZeRO stage 3 (aka FSDP) how it works (baby version)



Communication cost – 2 all gather (#param), 1 reduce-scatter (#param).

Actual picture of how FDSP / ZeRO stage 3 works

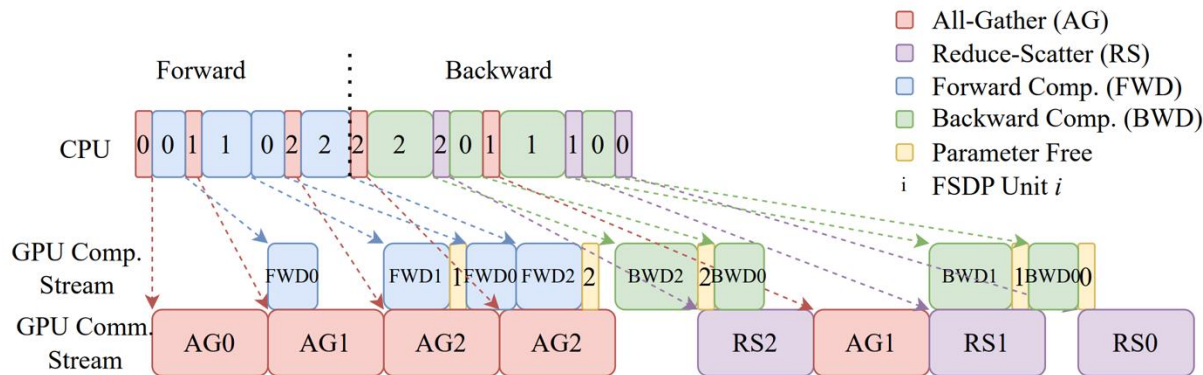
Let's walk through a FSDP example to see some important ideas

Incremental computation / communication

- Parameters / gradients are requested / sent and then immediately freed

Overlapping communication and computation ($W_1W_0 + W_2W_0$) $x = y$

- The all-gathers happen all at once while forward happens, masking the comm cost.



What's the point?

Distributed data parallel costs $2 \times \#$ param communication

What about ZeRO?

- **Zero stage 1** is $2 \times \#$ param – it's free! – you might as well always do it
- **Zero stage 2** is $2 \times \#$ param – it's (almost) free (ignoring overhead)
- **Zero stage 3** is $3 \times \#$ param – 1.5x comm cost, but that's not bad! (ignoring latency..)

This is also conceptually very simple – write a FSDP block wrapper.

ZeRO in practice: will it fit?

Pure BF16 training (with Kahan summation), is viable and optimizer states are less beefy. Let's say BF16 for everything but the master weights – 12 bytes per param

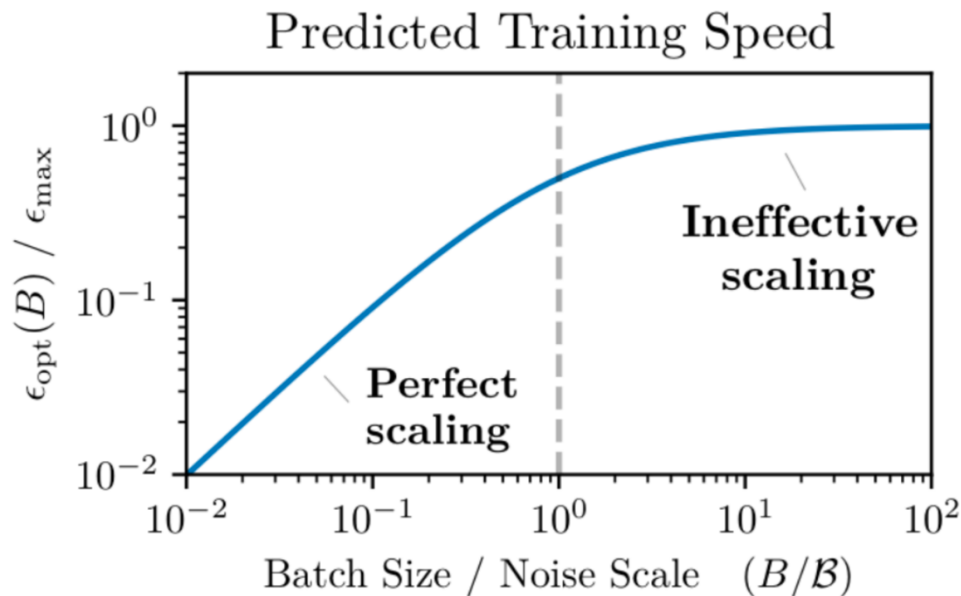
On a 8X A100 80G..

	Max size (params)	Formula for B/param
Baseline	6.66.. B	12
Zero stage 1	16 B	5
Zero stage 2	24.62 B	$2 (\text{param}) + (10 (\text{grad} + \text{state})) / 8$
Zero stage 3	53.33 B	12/8

Issues remain with data parallel – compute scaling

With data parallel, **#machines** < **batch size** (and near this, comm overhead is high)

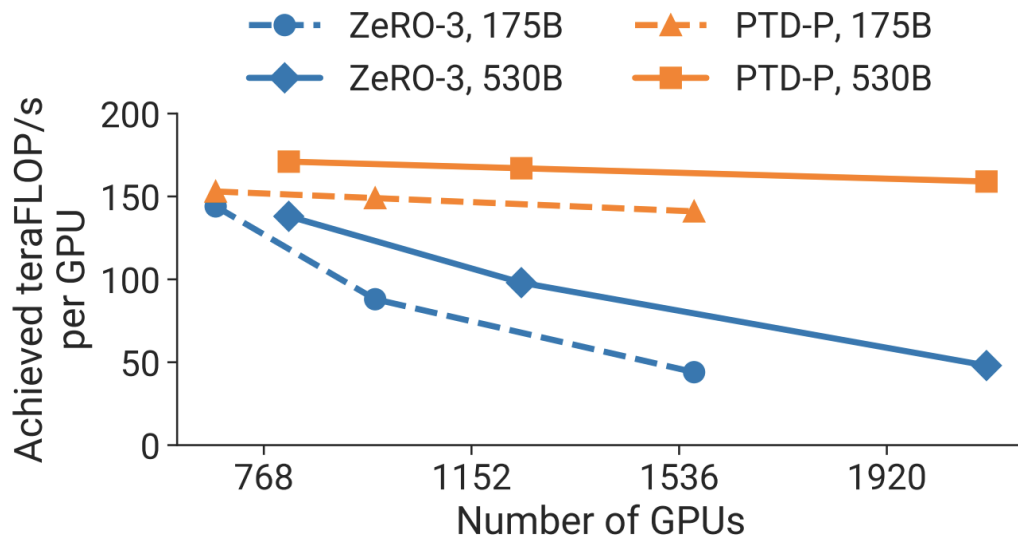
And there's diminishing returns to batch sizes



Issues remain with data parallel – models don't fit

Zero stages 1 and 2 don't let you scale memory

Zero stage 3 is nice in principle, but *does not reduce activation memory*



Better ways to split up the model is needed...

Beyond data parallel – model parallelism

Scaling up in memory (without changing batch size) with model parallelism

What model parallelism is..

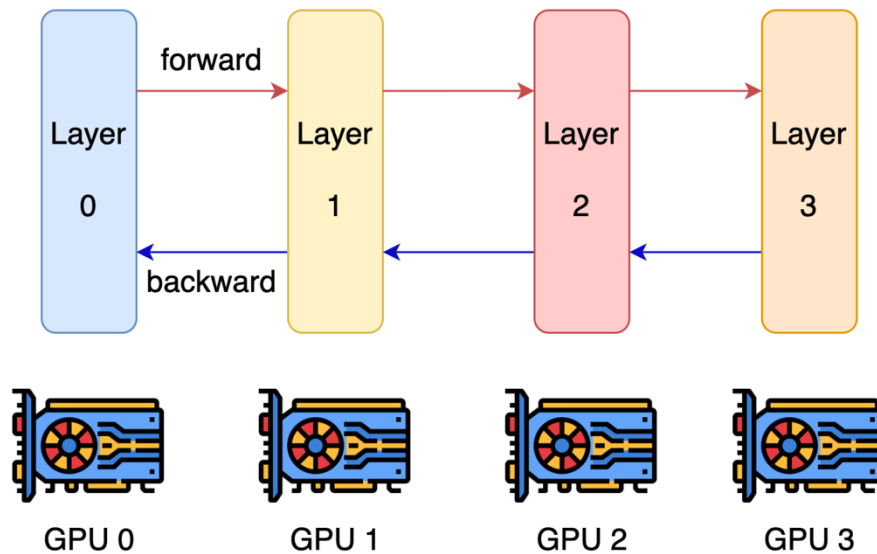
- It splits up the parameters across GPUs (like zero3)..
- But communicate activations (while zero3 sends params).

We cover three different types of parallelism

1. Pipeline parallel
2. Tensor parallel (+ Sequence parallel)
3. Expert parallel

These correspond to three different ways of cutting up the model.

Layer-wise parallel

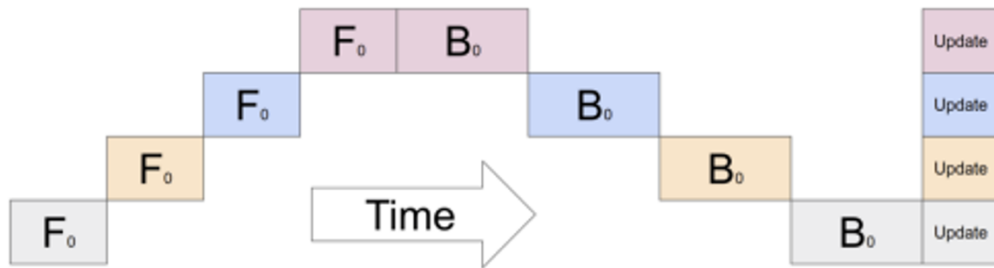


Layer-wise parallel cuts up layers, assigns some subset to GPUS.
Activations and partial gradients are passed back and forth

What's wrong with layer-wise parallel

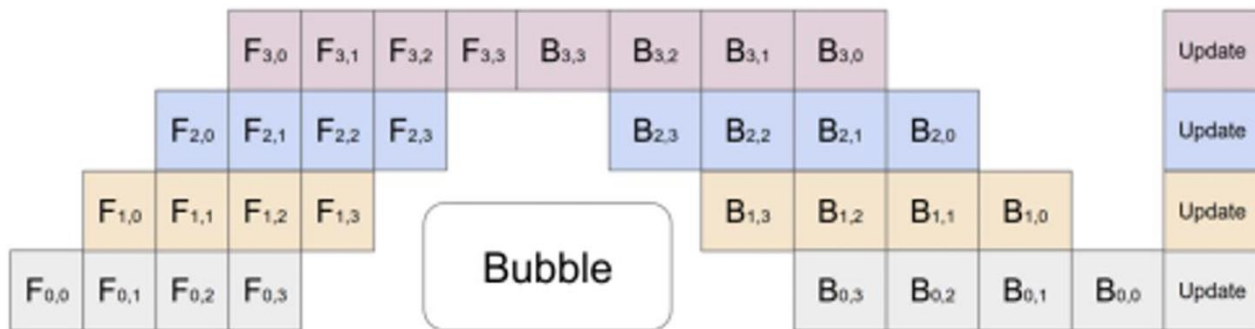
Utilization of layer-wise parallelism is *terrible*..

With n gpus, each gpu is active $\frac{1}{n}$ of the time.



Each GPU is idling most of the time, waiting for the backward pass to propagate back

A solution: pipeline parallel



Solution: Pipeline-parallel.

Process 'micro-batches' (in this case, 4).

Send off the first microbatch and start computing the second.

The ratio of bubble time to useful compute is .. $\frac{n_{stages}-1}{n_{micro}}$ so we need a big batch size!

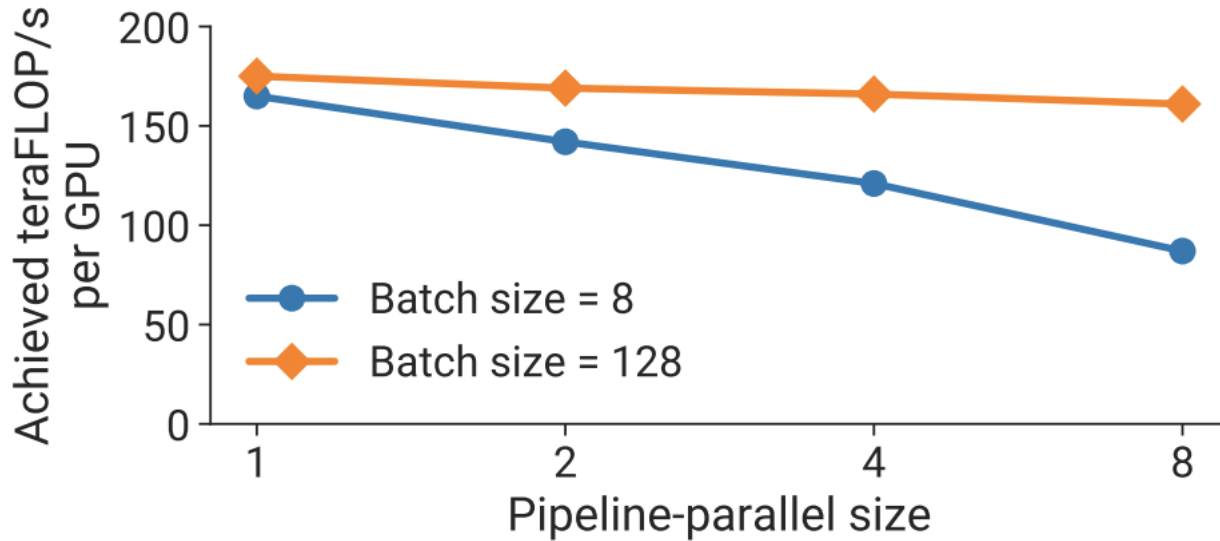
Why pipeline parallel?

Pipelines seem terrible. Why do we do it?

1. Pipelines save memory (compared to DDP)
2. Pipelines can have good communication properties (compared to FDSP) – it depends only on activations ($b \times s \times h$) and is *point to point*

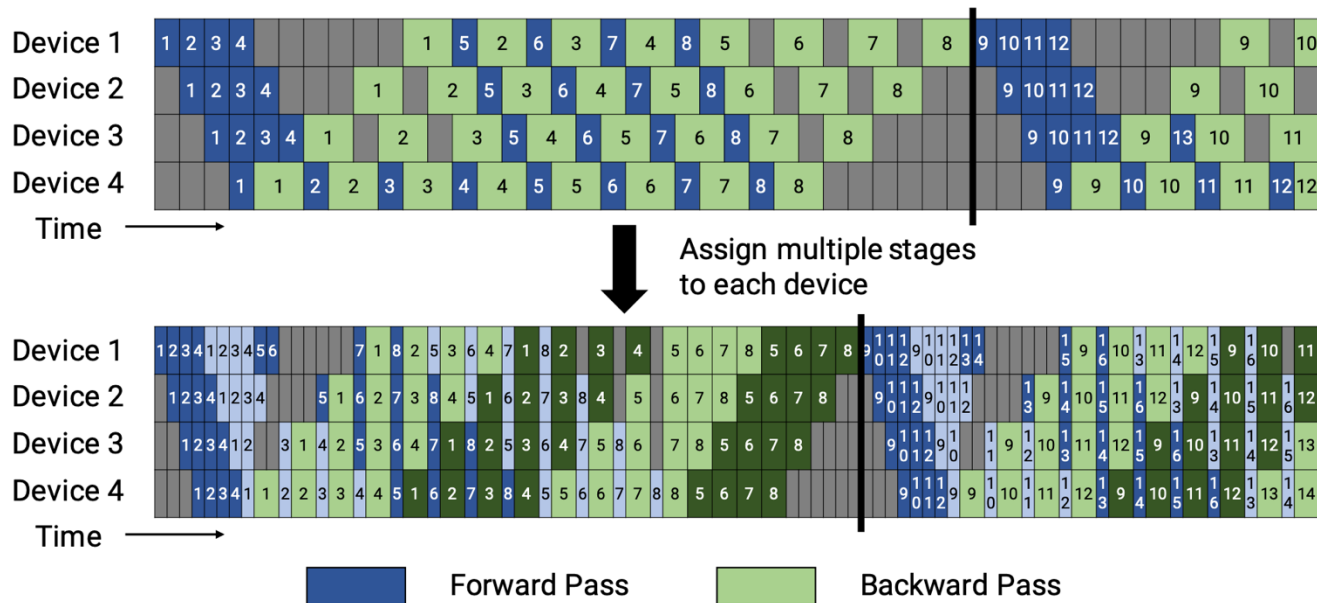
Generally, we will use pipelines on slower network links (i.e. inter-node) as a way to get better memory-wise scaling.

Pipeline performance is highly dependent on batch size



Batch sizes are key to hiding the bubble – otherwise pipeline rapidly degrades perf

Trading communication bandwidth for utilization



Some more crazy pipeline patterns can improve utilization, but at the cost of bandwidth

'Zero bubble' pipelining

Split up backwards into two parts

1. Backpropagating activations (z,x)
2. Computing weight gradients (W)

The second part can be done whenever

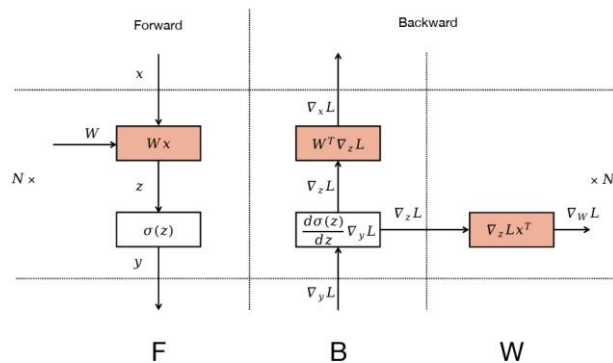


Figure 1: Computation Graph for MLP.



Figure 2: 1F1B pipeline schedule.

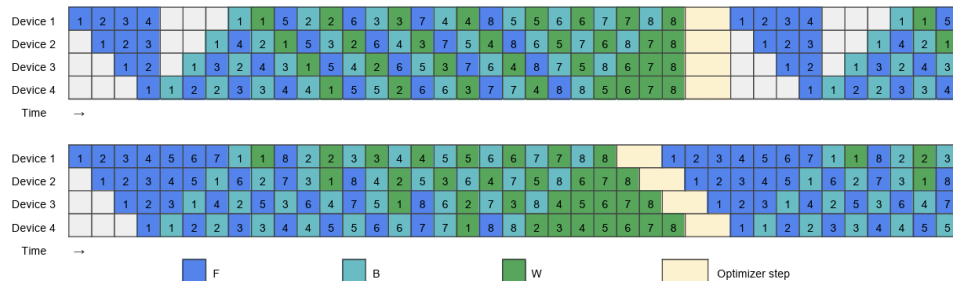
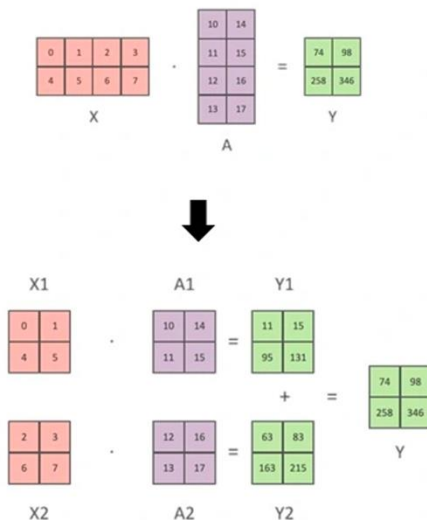


Figure 3: Handcrafted pipeline schedules, top: ZB-H1; bottom: ZB-H2

Model parallel along the width axes

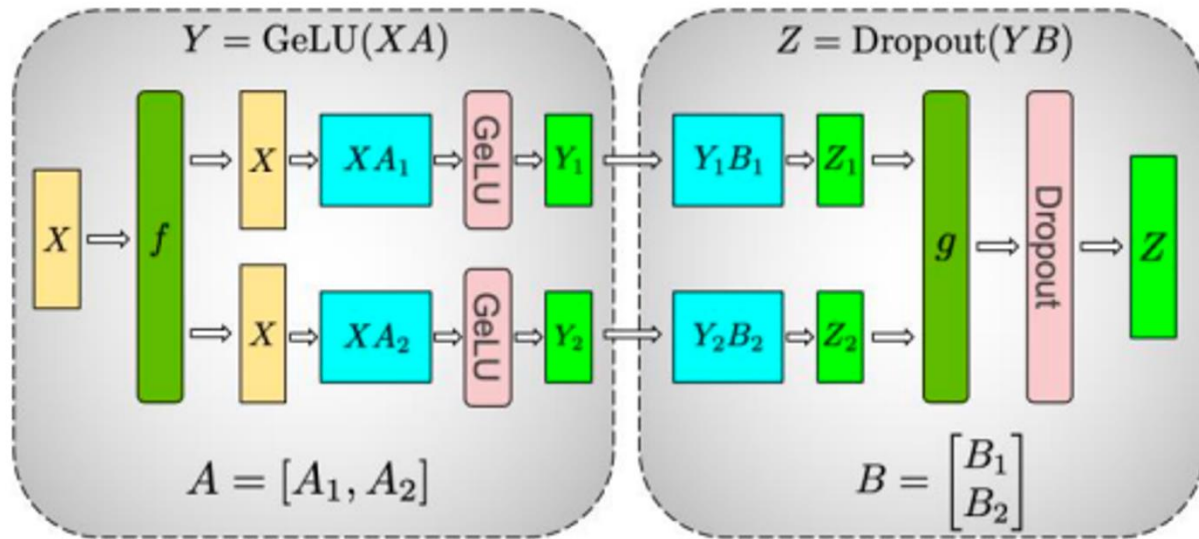
Are there model parallel schemes with better utilization?

We can think of pipeline parallel as cutting up along depth. What about width?



Simple matrix multiplication observation: decompose into submatrices, add partial sums

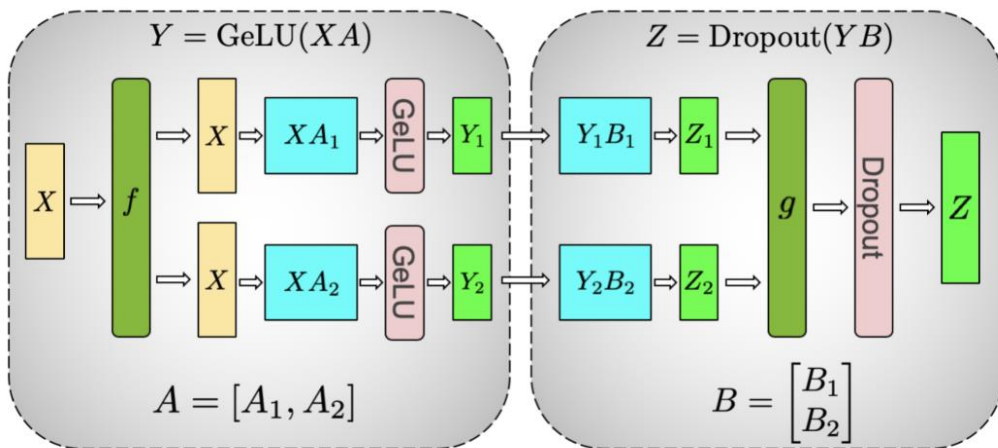
Tensor parallel – GPUs have submatrices



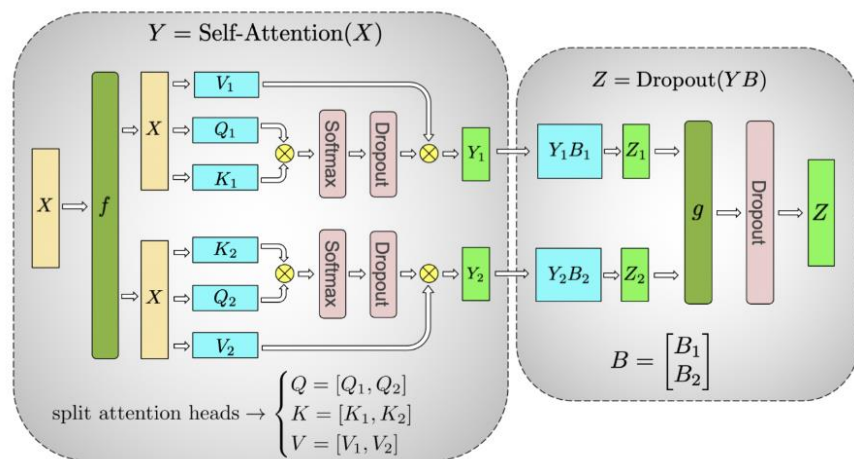
Assign columns (A_1, A_2) and rows (B_1, B_2) to separate GPUs.

- In the forward pass, f is the identity, and g is an all-reduce.
- In the backward pass, f is an all-reduce, g is the identity.

Row vs Column tensor parallel



(a) MLP



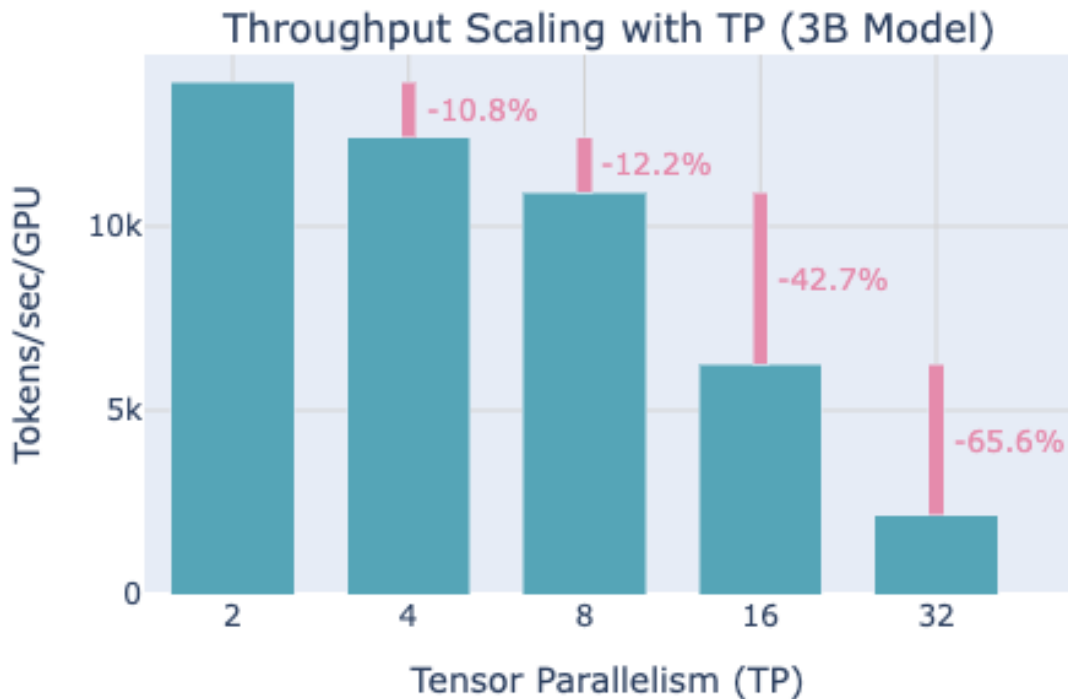
(b) Self-Attention

How do we split up in tensor parallel across a transformer block?

- Columnwise – QKV, up-projection
- Rowwise – Attention output, down-projection
- Replicated – norms, routers, etc

When do we tensor parallel?

On GPUs, tensor parallel within a node (up to 8 GPUs) due to high speed interconnects.



Tensor parallel – pros and cons vs pipeline parallel

How do things compare to pipeline parallel?

Pros – no bubble. If your network is fast enough, there's no waiting for others.

- low complexity – simple to 'wrap' models without major infra changes
- doesn't need large batch sizes to work well

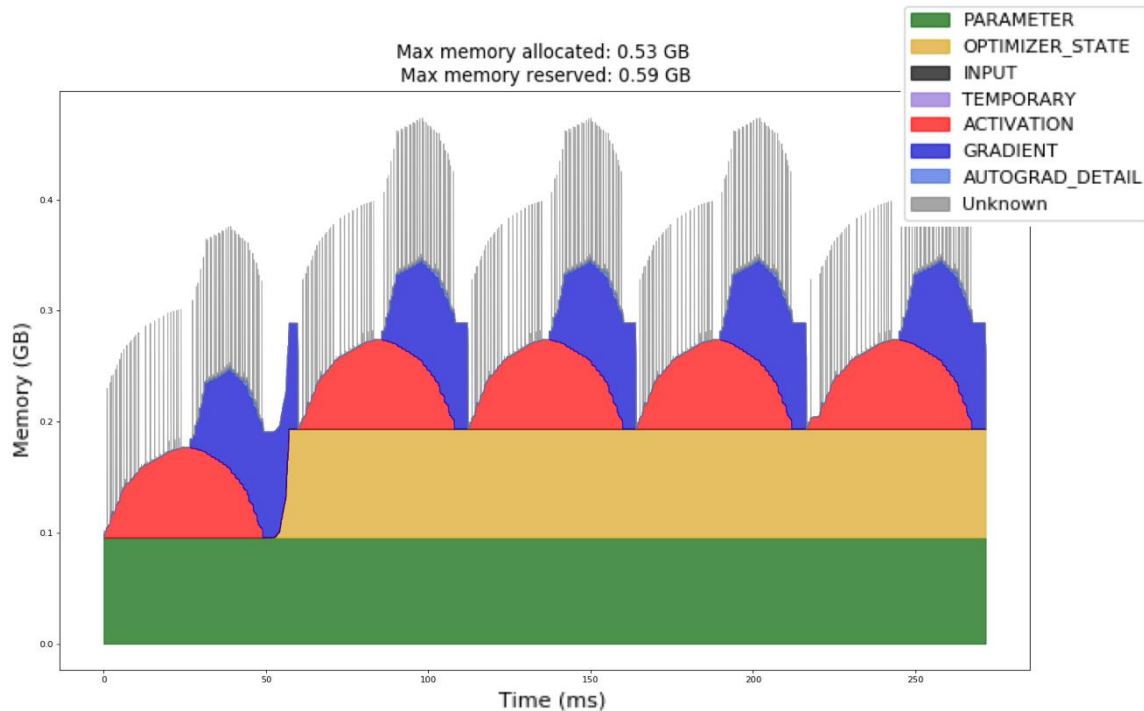
Cons – *much* larger communication than pipeline parallel.

- Pipeline: bsh point-to-point communication per microbatch
- Tensor: $8bsh \left(\frac{n_{devices}-1}{n_{devices}} \right)$ per layer and *all-reduce* communication.

Use tensor parallel whenever we have low-latency, high-bandwidth interconnects

A final complexity – memory is dynamic!

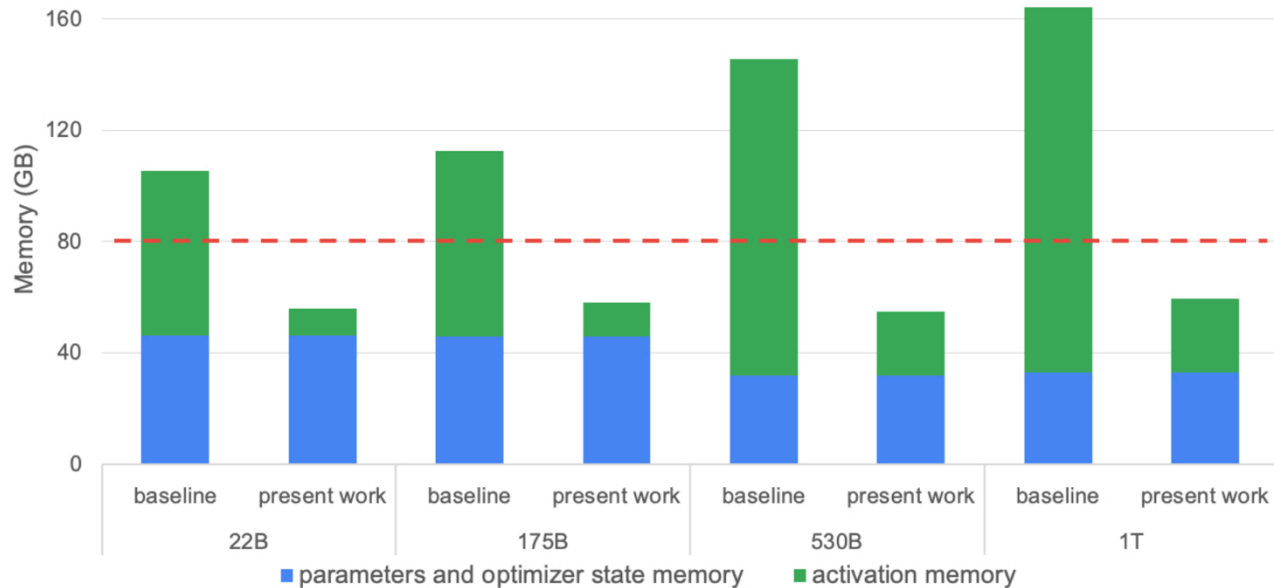
Memory isn't just the static bits, but also activations! This can be big



A final complexity – activation memory

Thus far, we have only really discussed parameter memory.

Tensor and pipeline parallel can linearly reduce those..**but what about activations?**



What's the activation memory per layer?

Starting point: activation memory needed if storing everything

$$\text{Activations memory per layer} = sbh \left(34 + 5 \frac{as}{h} \right).$$

- The $5 \frac{as}{h}$ terms come from the quadratic attention terms incl dropout
- As with flash attention, we can drop this term via recomputation

a	number of attention heads	p	pipeline parallel size
b	microbatch size	s	sequence length
h	hidden dimension size	t	tensor parallel size
L	number of transformer layers	v	vocabulary size

Activation under tensor parallel

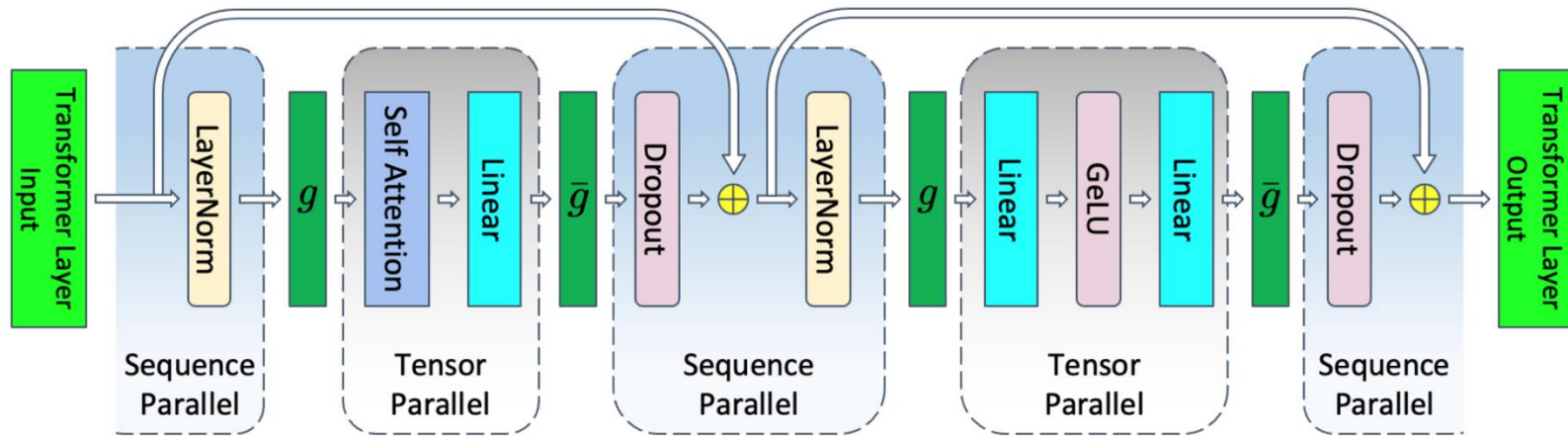
$$\underline{\text{Activations memory per layer}} = sbh \left(10 + \frac{24}{t} + 5 \frac{as}{ht} \right)$$

Tensor parallel splits out the matrix multiplies in attention + MLP

The remaining **10** term is for the LayerNorm (4sbh), Dropout (2sbh), and inputs to the attention and MLP (4sbh). These terms alone will continue to grow with size

a	number of attention heads	p	pipeline parallel size
b	microbatch size	s	sequence length
h	hidden dimension size	t	tensor parallel size
L	number of transformer layers	v	vocabulary size

Making memory truly linear – sequence parallel



Observation: all the 10sbh terms are pointwise ops over the sequence
... so split up the layer norm/dropout layers along the sequence axis.

- In the forward pass, ' g ' is an all gather, ' $\bar{g}$ ' is reduce-scatter
- In the backward pass, the two are reversed.

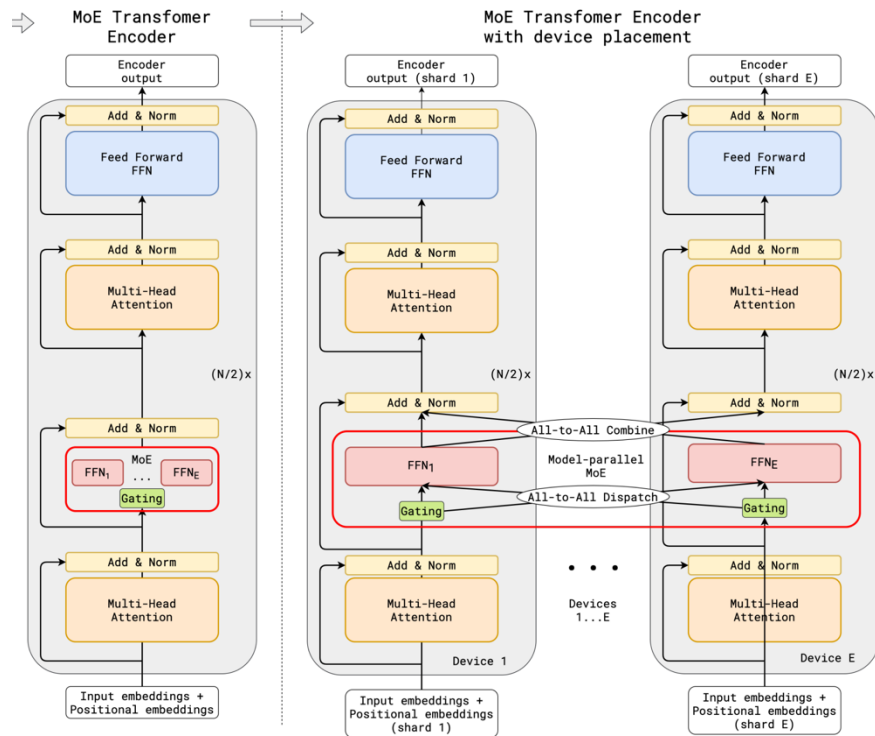
Making activation memory fully scale with more machines

Putting it together to get full linear scaling for memory.

Configuration	Activations Memory Per Transformer Layer
no parallelism	$sbh \left(34 + 5 \frac{as}{h} \right)$
tensor parallel (baseline)	$sbh \left(10 + \frac{24}{t} + 5 \frac{as}{ht} \right)$
tensor + sequence parallel	$sbh \left(\frac{34}{t} + 5 \frac{as}{ht} \right)$
tensor parallel + selective activation recomputation	$sbh \left(10 + \frac{24}{t} \right)$
tensor parallel + sequence parallel + selective activation recomputation	$sbh \left(\frac{34}{t} \right)$

Expert parallelism

Instead of splitting up the matmul, split up the experts and route activations



Why EP?

EP is *roughly* like TP in behavior for MLPs – high bandwidth, reduces activation

Guideline 4: Prefer EP over TP for Expert Layers

EP Advantages	Details
Better GEMM efficiency	Larger local matrix sizes improve GPU utilization
Lower communication	EP has less communication overhead than TP for MoE layers
Simpler computation graph	Easier to overlap communication with computation
Token permutation	When <code>EP = num_experts</code> , local token permutation is eliminated

Example: For Mixtral 8x7B, `EP8xTP1` outperforms `EP4xTP2`.

But, splitting matmuls can reduce efficiency vs routing activations

Complexity – combining EP and others

Naively – we can compose EP and all other strategies..

But there are important notes – DP usually shares replicas with EP splits (so EP<DP)

DP and TP can interact badly to lower utilization

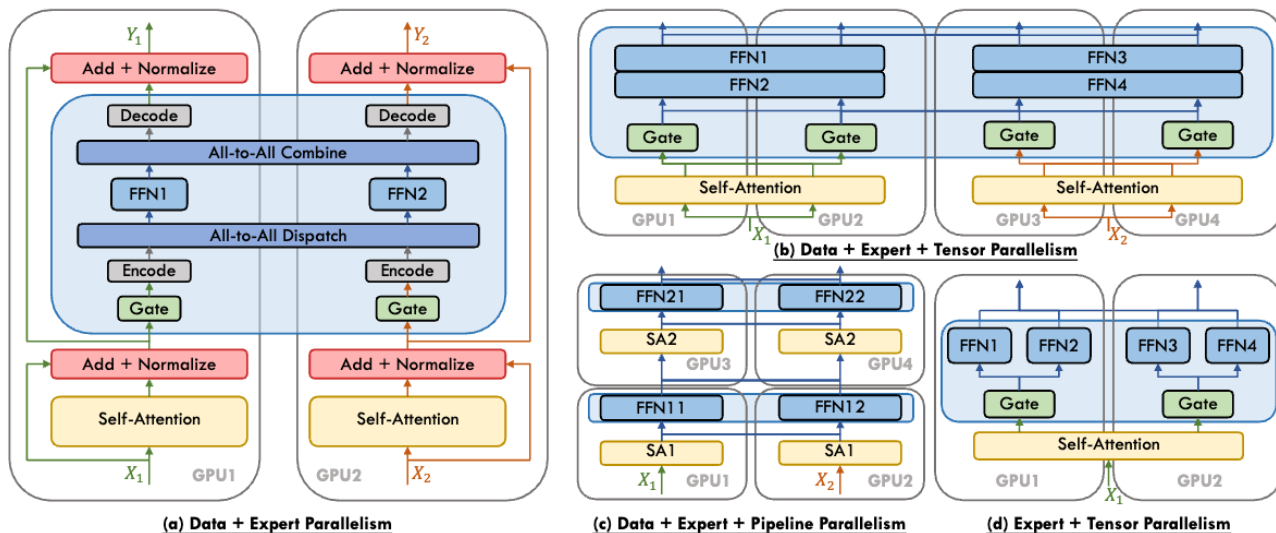


Fig. 8. Schematic depiction of diverse parallel strategies for MoE. For clarity and conciseness, this illustration omits some All-to-All, All-Reduce, Point-to-Point communication within parallelism, and Normalization, Encode, Decode, Gate in subfigures (b), (c), and (d).

Complexity: Decoupling attention / expert parallelism

Recall: MoEs apply to the MLPs, not the attention (with some exotic exceptions)

This can lead to some imbalance between attention and MLPs for parallelism

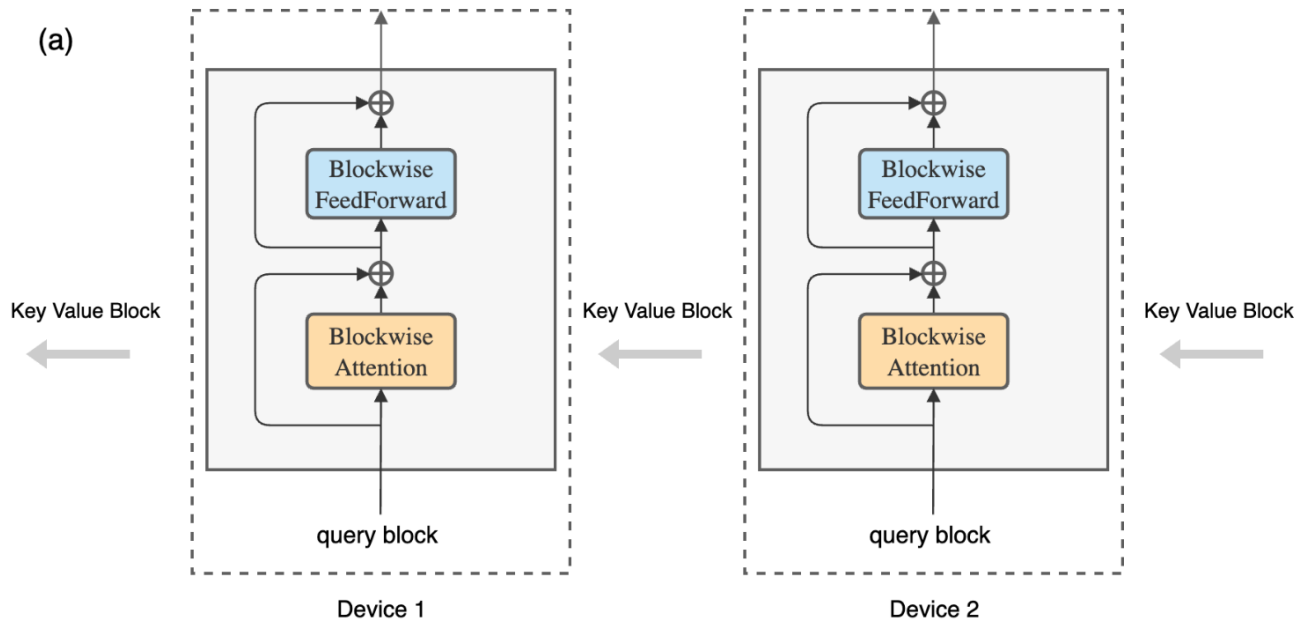
- High TP – useful for attention parallelization (since we can't use EP)
- Low TP – useful for MLP parallelization (we'd rather EP than TP)

What to do? More flexible parallelism strategies (i.e. in Megatron)

MoE Parallel Folding is Megatron Core's solution that decouples attention and MoE parallelism :		
Parallelism Group	Attention Layers	MoE Layers
Dimensions	$TP \times CP \times DP \times PP$	$ETP \times EP \times EDP \times PP$

Separates out TP/CP/DP (for attention) and (ETP/EP/EDP) for MLPs

Other parallelism strategies



Context parallel / Ring attention (split activations across GPUs in a long sequence)

Recap: LLM parallelism table

Method	Comm/Sync	Param memory per rank	Activation / KV memory per rank	Main bandwidth cost	Scales global batch?	Easy to use?
DDP / ZeRO-1	gradient all-reduce per step	No param scaling (ZeRO-1 only opt state)	None	Gradient traffic ~ $O(\text{params})$	Yes, linear in DP	Very
FSDP / ZeRO-3	Gradient all reduce – can be overlapped	~1/DP for params/grads/optimizer state	None	Param traffic ~ $O(\text{params})$, higher than DDP, overlap	Yes, linear in DP	Moderate
Pipeline parallel	Activation between layers, pipeline bubbles	~1/PP	Depends on pipeline buffers	Activation traffic between stages	No , but needs microbatches	Hard
Tensor parallel	Blocking activation communication	~1/TP for TP-sharded weights	~1/TP for relevant matmul activations w/ SP	Activation-sized collectives every block	No	Hard
Sequence / Context parallel	per-layer sequence-shard exchange	None	~1/SP or 1/CP on sequence-side activations / KV	Activation / KV communication	No	Hard
Expert parallel (MoE)	token dispatch - all-to-all per MoE	~1/EP for expert weights only	None	Token-routing all-to-all	No , but needs enough tokens per expert	Hard

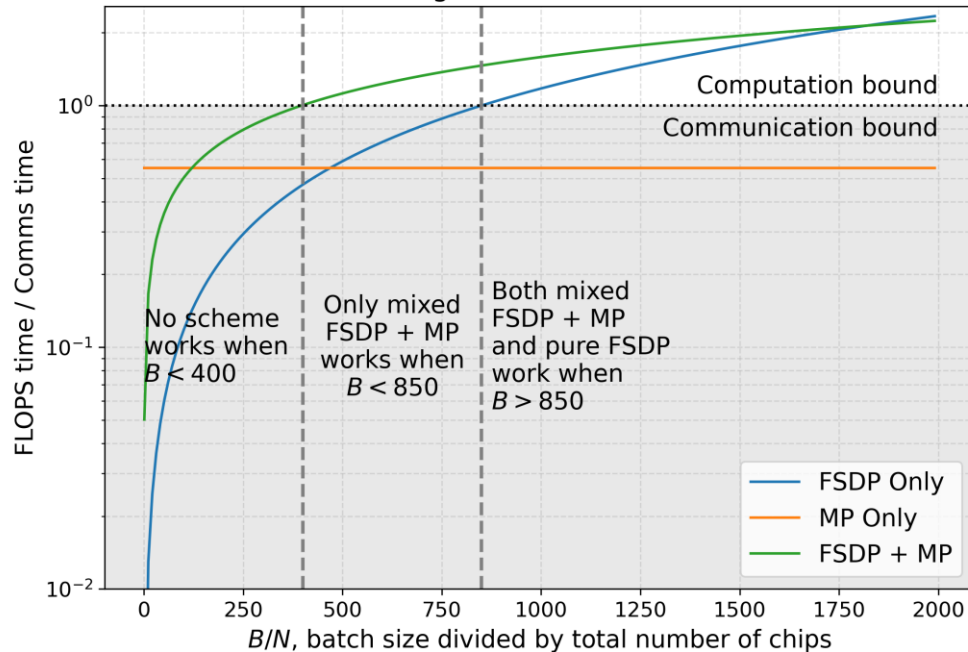
Model vs Tensor parallel (TPU book)

Key quantity

Global batch size (divided by GPU)

Strategy	Compute per layer (ignoring gating einsum)	Comms per layer (bytes, forward + backward pass)
DP	$4BDF/X + 8BDF/X$	$0 + 8DF$
FSDP	$4BDF/X + 8BDF/X$	$4DF + 8DF$
MP	$4BDF/Y + 8BDF/Y$	$4BD + 4BD$
FSDP + MP	$4BDF/(XY) + 8BDF/(XY)$	$(4BD/X + 4DF/Y) + (8BD/X + 8DF/Y)$

Batch-size scaling behavior of parallelization strategies on a 4x4x4 mesh

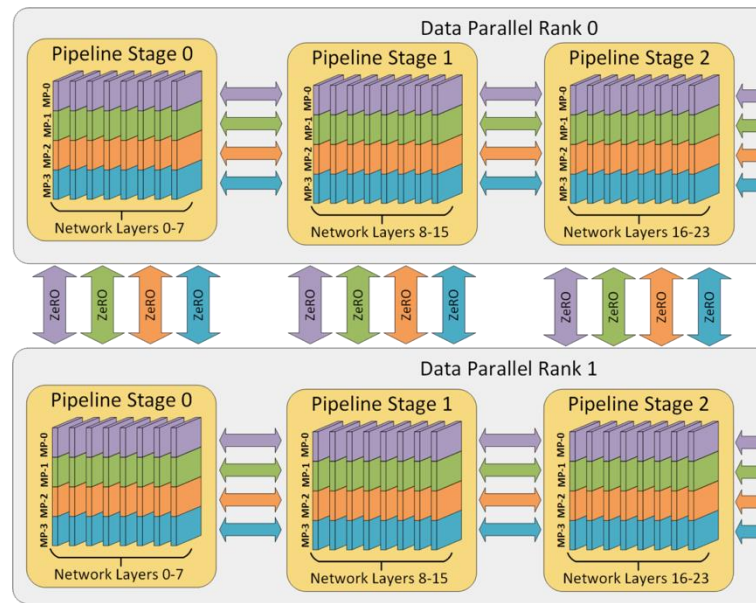


'3D (4D) parallelism' – putting it all together

Simple rules of thumb from the literature.

1. Until your model fits in memory..
 - Tensor / expert parallel up to GPUs / machine
 - Pipeline parallel across machines
 - › (Or use Zero-3, depending on BW)
2. Then until you run out of GPUs
 - Scale the rest of the way with data parallel

If your batch size is small.. gradient accumulate to trade higher batch sizes for better communication efficiency.



Example from Megatron's current recommendations

The “basics”

Guideline 1: Minimize Model Parallelism, Maximize Data Parallelism

Aspect	Recommendation
Goal	Keep TP/EP/PP as small as possible while avoiding OOM
Why	Model parallelism introduces communication overhead that hurts performance
How	Use distributed optimizer (<code>--use-distributed-optimizer</code>) to shard optimizer states across DP ranks, freeing memory for larger DP size

Guideline 2: Keep EP and TP Communication Within NVLink Domain

Aspect	Recommendation
Goal	Ensure EP×TP fits within a single node (typically 8 GPUs)
Why	EP and TP are communication-intensive; NVLink provides much higher bandwidth than cross-node interconnects
Scaling	When scaling beyond one node, prefer PP over expanding TP/EP across nodes

Note: For very large MoE models like DeepSeek-V3, the EP communication may exceed the NVLink bandwidth. In this case, consider using 1F1B A2A Overlap to overlap the EP communication.

Guideline 3: Use Pipeline Parallelism (PP) for Multi-Node Scaling

Aspect	Recommendation
Goal	Use PP to distribute layers across nodes while keeping EP×TP within NVLink
VPP	Enable Virtual Pipeline Parallelism to reduce pipeline bubbles when <code>PP ≥ 2</code>
Config	Set <code>--num-layers-per-virtual-pipeline-stage</code> to control VPP size

Plus more advanced “extras”

Guideline 4: Prefer EP over TP for Expert Layers

EP Advantages	Details
Better GEMM efficiency	Larger local matrix sizes improve GPU utilization
Lower communication	EP has less communication overhead than TP for MoE layers
Simpler computation graph	Easier to overlap communication with computation
Token permutation	When <code>EP = num_experts</code> , local token permutation is eliminated

Example: For Mixtral 8x7B, `EP8×TP1` outperforms `EP4×TP2`.

Guideline 5: Enable Context Parallelism (CP) for Long Sequences

Aspect	Recommendation
When to use	Sequence length $\geq$ 8K tokens
Key factor	CP efficiency depends on overlapping communication with computation
Config	Set <code>--context-parallel-size</code> to partition sequences across GPUs

Scaling strategies from Narayanan 2021

Efficient Large-Scale Language Model Training on GPU Clusters Using Megatron-LM

Deepak Narayanan^{1*}, Mohammad Shoeybi¹, Jared Casper¹, Patrick LeGresley¹,
Mostofa Patwary¹, Vijay Korthikanti¹, Dmitri Vainbrand¹, Prethvi Kashinkunti¹,
Julie Bernauer², Bryan Catanzaro¹, Amar Phanishayee², Matei Zaharia²
¹NVIDIA ²Stanford University ^{*}Microsoft Research

Number of parameters (billion)	Attention heads	Hidden size	Number of layers	Tensor model-parallel size	Pipeline model-parallel size	Number of GPUs	Batch size	Achieved teraFLOP/s per GPU	Percentage of theoretical peak FLOP/s	Achieved aggregate petaFLOP/s	DP size
1.7	24	2304	24	1	1	32	512	137	44%	4.4	32
3.6	32	3072	30	2	1	64	512	138	44%	8.8	32
7.5	32	4096	36	4	1	128	512	142	46%	18.2	32
18.4	48	6144	40	8	1	256	1024	135	43%	34.6	32
39.1	64	8192	48	8	2	512	1536	138	44%	70.8	32
76.1	80	10240	60	8	4	1024	1792	140	45%	143.8	32
145.6	96	12288	80	8	8	1536	2304	148	47%	227.1	24
310.1	128	16384	96	8	16	1920	2160	155	50%	297.4	15
529.6	128	20480	105	8	35	2520	2520	163	52%	410.2	9
1008.0	160	25600	128	8	64	3072	3072	163	52%	502.0	6

Notes

- Tensor parallel first up to 8, then caps out at 8.
- Pipeline parallel goes up to make the model fit.
- Data parallel gradually decreases with scale, with the largest model having DP=6

Careful '3D' parallelism gives linear gains

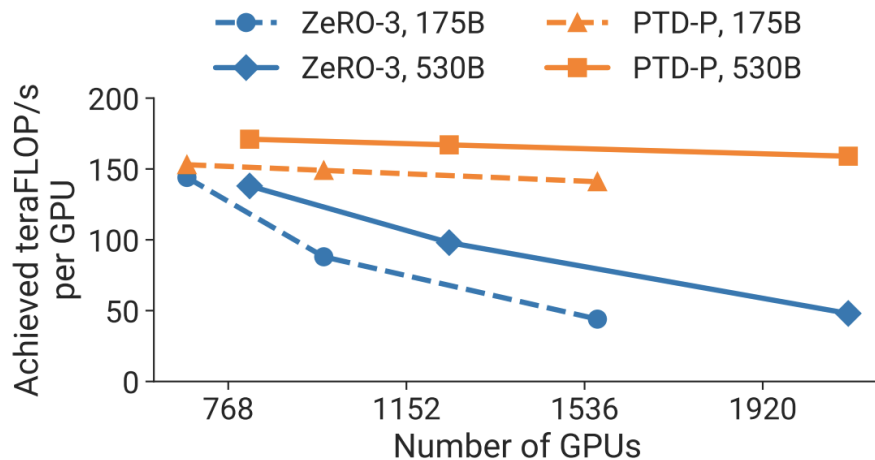


Figure 10: Throughput per GPU of PTD-P and ZeRO-3 for two different GPT models (the 175B GPT-3 model is shown with dotted lines, and the 530B model is shown with solid lines). Global batch sizes are fixed and ZeRO-3 is used without any model parallelism.

More GPUS, same, flat utilization!

Tensor parallel = 8 is often optimal

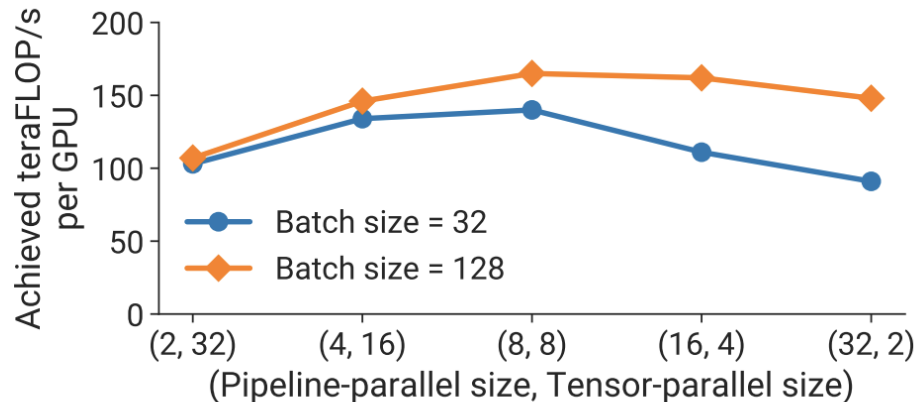
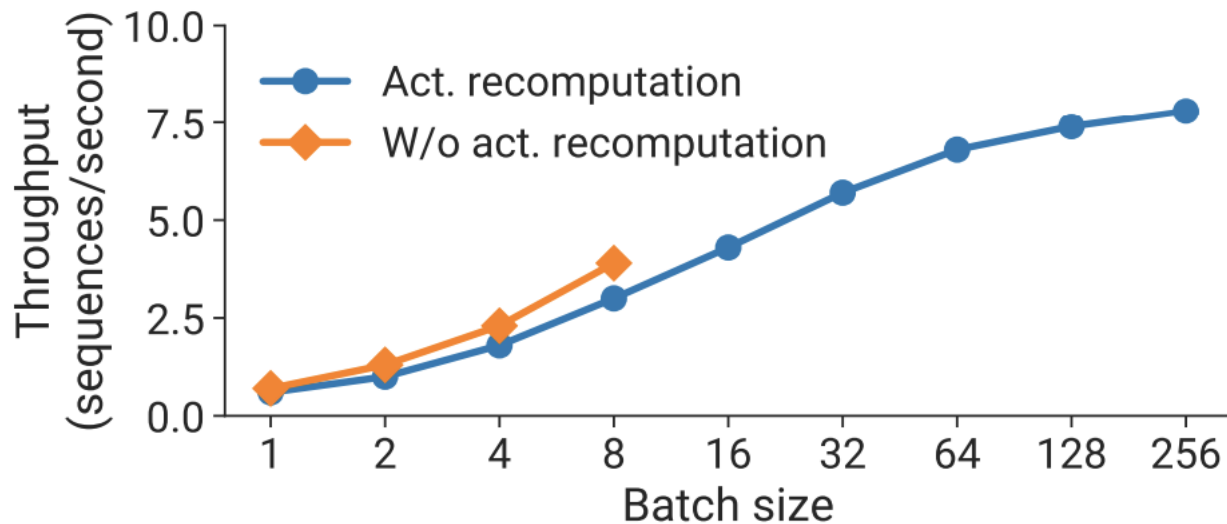


Figure 13: Throughput per GPU of various parallel configurations that combine pipeline and tensor model parallelism using a GPT model with 162.2 billion parameters and 64 A100 GPUs.

When parallelizing across 64 machines – it's best to use a 8 x 8 configuration.

Activation recomputation can pay for itself (via memory)



Activation recomputation enables larger batches, improving throughput (t=8, p=16)

Recent LMs – what do they do?

3.1 Distributed Training Framework

We train our models using the *ZeRO* optimizer strategy (Rajbhandari et al., 2019) via PyTorch's FSDP framework (Zhao et al., 2023), which reduces memory consumption by sharding the model weights and their corresponding optimizer state across GPUs. At the 7B scale, this enables training with a micro-batch size of 4096 tokens per GPU on our hardware (see Section 3.4). For OLMo-1B and -7B models, we use a constant global batch size of approximately 4M tokens (2048 instances, each with a sequence length of 2048 tokens). For OLMo-65B model (currently training), we use a batch size warmup that starts at approximately 2M tokens (1024 instances), then doubles every 100B tokens until reaching approximately 16M tokens (8192 instances).

To improve throughput, we employ mixed-precision training (Micikevicius et al., 2017) through FSDP's built-in settings and PyTorch's amp module. The latter ensures that certain operations like the softmax always run in full precision to improve stability, while all other operations run in half-precision with the bfloat16 format. Under our specific settings, the sharded model weights and optimizer state local to each GPU are kept in full precision. The weights within each transformer block are only cast to bfloat16 when the full-sized parameters are materialized on each GPU during the forward and backward passes. Gradients are reduced across GPUs in full precision.

Dolma – 7B model, FDSP (probably fits intra-node)

DeepSeek

2.4. Infrastructures

We use an efficient and light-weight training framework named HAI-LLM (High-flyer, 2023) to train and evaluate large language models. Data parallelism, tensor parallelism, sequence parallelism, and 1F1B pipeline parallelism are integrated into this framework as done in Megatron (Korthikanti et al., 2023; Narayanan et al., 2021; Shoeybi et al., 2019). We also leverage the flash attention (Dao, 2023; Dao et al., 2022) technique to improve hardware utilization. ZeRO-1 (Rajbhandari et al., 2020) is exploited to partition optimizer states over data parallel ranks. Efforts are also made to overlap computation and communication to minimize additional waiting overhead, including the backward procedure of the last micro-batch and reduce-scatter operation in ZeRO-1, and GEMM computation and all-gather/reduce-scatter in sequence parallel. Some layers/operators are fused to speed up training, including LayerNorm, GEMM whenever possible, and Adam updates. To improve model training stability, we train the model in bf16 precision but accumulate gradients in fp32 precision. In-place cross-entropy is performed to reduce GPU memory consumption, i.e.: we convert bf16 logits to fp32 precision on the fly in the cross-entropy CUDA kernel (instead of converting it beforehand in HBM), calculate the corresponding bf16 gradient, and overwrite logits with its gradient.

DeepSeek – ZeRO stage 1 with Tensor, Sequence, and Pipeline parallel

V3 – PP (16), EP (64-way, 8 nodes), ZeRO stage 1

EP uses 1F1B A2A Overlap

Performance and Cost Efficiency

Memory and communication restrictions are the two major technical challenges of large scale model training requiring integrated solutions beyond adding more GPUs. We use and improve upon the following techniques to tackle the memory and communication restrictions: (1) ZeRO-1 [60] to remove the memory consumption by partitioning optimizer states cross data-parallel processes; (2) tensor parallel combined with pipeline parallel [70] within each compute node to avoid inter-node communication bottleneck, and the 3D parallel strategy is well designed and optimized to avoid using activation checkpointing and minimize the pipeline bubbles; (3) kernel fusion techniques like flash attention [15] [14] and JIT kernels to reduce redundant global memory access and consumption; (4) topology-aware resource allocation (ranking strategy) to minimize the communication across different layers of switches, which is the limitation of a typical fat-tree-topology.

Yi - ZeRO stage 1 + Tensor + Pipeline parallel

Yi-lightning (2025) – Tensor replaced by Expert parallelism

Llama3 405B

GPUs	TP	CP	PP	DP	Seq. Len.	Batch size/DP	Tokens/Batch	TFLOPs/GPU	BF16 MFU
8,192	8	1	16	64	8,192	32	16M	430	43%
16,384	8	1	16	128	8,192	16	16M	400	41%
16,384	8	16	16	8	131,072	16	16M	380	38%

(Stage 1, small bsz training, Stage 2 pretraining, Stage 3 long-context)

Network-aware parallelism configuration. The order of parallelism dimensions, [TP, CP, PP, DP], is optimized for network communication. The innermost parallelism requires the highest network bandwidth and lowest latency, and hence is usually constrained to within the same server. The outermost parallelism may spread across a multi-hop network and should tolerate higher network latency. Therefore, based on the requirements for network bandwidth and latency, we place parallelism dimensions in the order of [TP, CP, PP, DP]. DP (*i.e.*, FSDP) is the outermost parallelism because it can tolerate longer network latency by asynchronously prefetching sharded model weights and reducing gradients. Identifying the optimal parallelism configuration with minimal communication overhead while avoiding GPU memory overflow is challenging. We develop a memory consumption estimator and a performance-projection tool which helped us explore various parallelism configurations and project overall training performance and identify memory gaps effectively.

Llama 3 405B

Side note – Lots of GPU failures at this scale!

Component	Category	Interruption Count	% of Interruptions
Faulty GPU	GPU	148	30.1%
GPU HBM3 Memory	GPU	72	17.2%
Software Bug	Dependency	54	12.9%
Network Switch/Cable	Network	35	8.4%
Host Maintenance	Unplanned Maintenance	32	7.6%
GPU SRAM Memory	GPU	19	4.5%
GPU System Processor	GPU	17	4.1%
NIC	Host	7	1.7%
NCCL Watchdog Timeouts	Unknown	7	1.7%
Silent Data Corruption	GPU	6	1.4%
GPU Thermal Interface + Sensor	GPU	6	1.4%
SSD	Host	3	0.7%
Power Supply	Host	3	0.7%
Server Chassis	Host	2	0.5%
IO Expansion Board	Host	2	0.5%
Dependency	Dependency	2	0.5%
CPU	Host	2	0.5%
System Memory	Host	2	0.5%

Table 5 Root-cause categorization of unexpected interruptions during a 54-day period of Llama 3 405B pre-training. About 78% of unexpected interruptions were attributed to confirmed or suspected hardware issues.

Gemma 2

For 2, 9, 27B models

ZeRO-3, MP (=TP+SP), DP

3.3. Compute Infrastructure

We train our models with TPUv4, TPUv5e, and TPUv5p as outlined in [Table 3](#). For the 2B model, we train on a 2x16x16 configuration of TPUv5e, totaling 512 chips, with 512-way data replication and 1-way model sharding. For the 9B model, we train on an 8x16x32 configuration of TPUv4, totaling 4096 chips, with 1024-way data replication and 4-way model sharding. For the 27B model, we train on an 8x24x32 configuration of TPUv5p, totaling 6144 chips, with 768-way data replication and 8-way model sharding.

The optimizer state is further sharded using techniques similar to ZeRO-3 ([Ren et al., 2021](#)). For scales beyond a single pod, we perform a data-replica reduction over the data center network, using the Pathways approach of [Barham et al. \(2022\)](#). We also use the 'single controller' programming paradigm of Jax ([Roberts et al., 2023](#)) and Pathways ([Barham et al., 2022](#)). As in Gemma 1, we use the GSPMD partitioner ([Xu et al., 2021](#)) for training step computation and the MegaScale XLA compiler ([XLA, 2019](#)).

Mixtral 8x22B

From megatron docs –

TP / PP / CP / EP (4/4/1/8) – DP likely 2 (to add up to 256GPUs)

Mixture-of-Experts Models							
Model	Size	GPUs	TP	PP	CP	EP	Configuration Notes
Mixtral	8x7B	64	1	4	1	8	EP=8 for 8 experts
Mixtral	8x22B	256	4	4	1	8	TP+PP+EP for large MoE
DeepSeek-V3	671B	1024	2	16	1	64	Massive MoE with 256 experts

Nemotron 3 Super 120B-A12B

From the long-context extension section

Similar to Nemotron 3 Nano, we added a long-context phase (LC-Phase) at the end of pretraining. In the LC-Phase, we performed continuous pretraining (CPT) to equip the base model with long-context ability. We used a constant learning rate of $4.5 * 10^{-6}$ and global batch size of 16. We used 64-way context parallelism, 2-way tensor parallelism, and 64-way expert parallelism to train on GB200 GPUs. We reused the long-context document QA dataset from Nemotron 2 & 3 Nano. We allocated the document QA data to 20% in the Phase LC data blend, with the remaining 80% being downsampled Phase 2 data. We initially performed CPT on 1,048,576 (1m) context length. Such stage lasted for 34 billion tokens. Following that we added another stage to alternately train on both 1m and 4k sequences in order to mitigate the minor impact we observed on the math-related benchmarks. The second stage lasted for 17 billion tokens.

TP / PP / CP / EP (2/0/64/64)

Qwen 3

225B-A22B and 30B-A3B

Parallelism Configuration

Model	Mode	TP	PP	EP	Total GPUs	Use Case
Qwen3-30B-A3B	Pretrain	1	1	8	8	Pre-training (single node)
Qwen3-30B-A3B	Full SFT	1	1	8	8	Full supervised finetuning
Qwen3-30B-A3B	LoRA/DoRA	1	1	8	8	PEFT finetuning (single node)
Qwen3-235B-A22B	Pretrain	2	8	32	512	Pre-training (64 nodes)
Qwen3-235B-A22B	Full SFT	2	8	32	512	Full supervised finetuning (64 nodes)
Qwen3-235B-A22B	LoRA/DoRA	2	8	32	512	PEFT finetuning (64 nodes)

Workload family	Hardware	Typical band	Representative shape
DSV3, large-scale	H100	low-to-mid hundreds TFLOPS/GPU, high-teens MFU	TP2, EP64, PP8, DeepEP
DSV3, large-scale	B200	high-hundreds TFLOPS/GPU, mid-teens MFU	TP1, EP32, PP8, DeepEP
DSV3, large-scale	GB200	around 1K TFLOPS/GPU, low-20s MFU	TP1, EP64, PP4, HybridEP
DSV3, large-scale	GB300	above the GB200 band, often mid-20s MFU	TP1, EP64, PP4, HybridEP
Qwen3 235B	H100	low-300s TFLOPS/GPU, around 30% MFU	TP2, EP32, PP8, DeepEP
Qwen3 235B	GB200	high-hundreds TFLOPS/GPU in tuned runs	TP1 or TP2, EP32-64, PP4, HybridEP
Qwen3 30B	H100	low-200s TFLOPS/GPU	TP1, EP8, PP1, DeepEP
Qwen3-Next 80B	GB200	low-300s TFLOPS/GPU in BF16-class runs	TP1, EP32, PP2, HybridEP

Primarily expert parallel up to 8 GPUs, 2/8/32 parallelism for larger models

Overview table of model parallelism

	DP	TP/SP	EP	PP	CP
Deepseek	?? (Zero1)	1	8	16	??
Deepseek v3	?? (Zero1)	1	64	16	??
Yi	?? (Zero1)	>0	1	>0	??
Llama3 405B	128	8	0	16	1
Gemma 2	768	8	0	0	0
Mixtral 8x22 (megatron)	2	4	8	4	1
Nemotron 3 120B (long context)	??	2	64	??	64
Qwen 3 (megatron)	??	2	32	8	1

Patterns – TP generally ≤ 8 . EP can be big (but hard!). Long context phases use large CP

Recap for the whole lecture

- ❖ Scaling beyond a certain point requires multi-gpu, multi-node parallelism
- ❖ No single solution to the parallelism problem (probably want all 3 approaches)
- ❖ Simple, interpretable rules of thumb for combining different forms of parallelism